

Name : Sk.Arshad Basha

Regno : 192312336

Compare different types Classification Algorithms and evaluate their performance.

CODE :

```
import numpy as np

# Dataset
X = np.array([
    [1, 1], [2, 1], [1, 2],
    [8, 8], [9, 8], [8, 9]
])
y = np.array([0, 0, 0, 1, 1, 1])

# Test data
test = np.array([2, 2])

# KNN
distance = np.sqrt(np.sum((X - test) ** 2, axis=1))
k = 3
nearest = np.argsort(distance)[:k]
prediction = np.bincount(y[nearest]).argmax()
print("KNN Prediction:", prediction)

# Accuracy
correct = 0
for i in range(len(X)):
    d = np.sqrt(np.sum((X - X[i]) ** 2, axis=1))
    d[i] = np.inf
    n = np.argsort(d)[:k]
    p = np.bincount(y[n]).argmax()
    if p == y[i]:
        correct += 1
accuracy = correct / len(X)

print("KNN Accuracy:", accuracy)
```

OUTPUT

```
KNN Prediction: 0
```

```
KNN Accuracy: 1.0
```

```
=== Code Execution Successful ===
```